

DeepSeek R1

Module 2 — Reading the paper

arxiv.org/pdf/2501.12948

Why this paper matters

Open weights + an open recipe for o1-level reasoning.

How do you train a model to *think*?

You don't write the reasoning for it. You **incentivize** it:

- Show a few examples of **thinking through** a problem (cold-start)
- Then **reward** the behavior you want with RL:
 - did it get the answer **right**? (accuracy)
 - did it use the **<think>** markers? (format)
 - did it **stay in one language**? (consistency)

No human writes the chains. The model discovers them.

The trick: verifiable rewards

For math and code you can **check the answer automatically** — no learned reward model, no human rater in the loop.

- Math → compare to the parsed ground-truth number
- Code → run the **unit tests**

A cheap, unhackable signal is what makes pure-RL reasoning possible.

R1-Zero vs R1

- **R1-Zero:** pure RL on a base model, no SFT. Works — but ugly outputs.
- **R1:** add a small cold-start SFT + a second SFT pass. Same skill, readable.

The "aha moment"

Table 2 | An interesting “aha moment” of an intermediate version of DeepSeek-R1-Zero. The model learns to rethink using an anthropomorphic tone. This is also an aha moment for us, allowing us to witness the power and beauty of reinforcement learning.

Question: If $a > 1$, then the sum of the real solutions of $\sqrt{a - \sqrt{a + x}} = x$ is equal to

Response: <think>

To solve the equation $\sqrt{a - \sqrt{a + x}} = x$, let's start by squaring both ...

$$\left(\sqrt{a - \sqrt{a + x}}\right)^2 = x^2 \implies a - \sqrt{a + x} = x^2.$$

Rearrange to isolate the inner square root term:

$$(a - x^2)^2 = a + x \implies a^2 - 2ax^2 + (x^2)^2 = a + x \implies x^4 - 2ax^2 - x + (a^2 - a) = 0$$

...

Wait, wait. Wait. That's an aha moment I can flag here.

Let's reevaluate this step-by-step to identify if the correct sum can be ...

We started with the equation:

$$\sqrt{a - \sqrt{a + x}} = x$$

First, let's square both sides:

$$a - \sqrt{a + x} = x^2 \implies \sqrt{a + x} = a - x^2$$

Next, I could square both sides again, treating the equation: ...

...

Accuracy curve

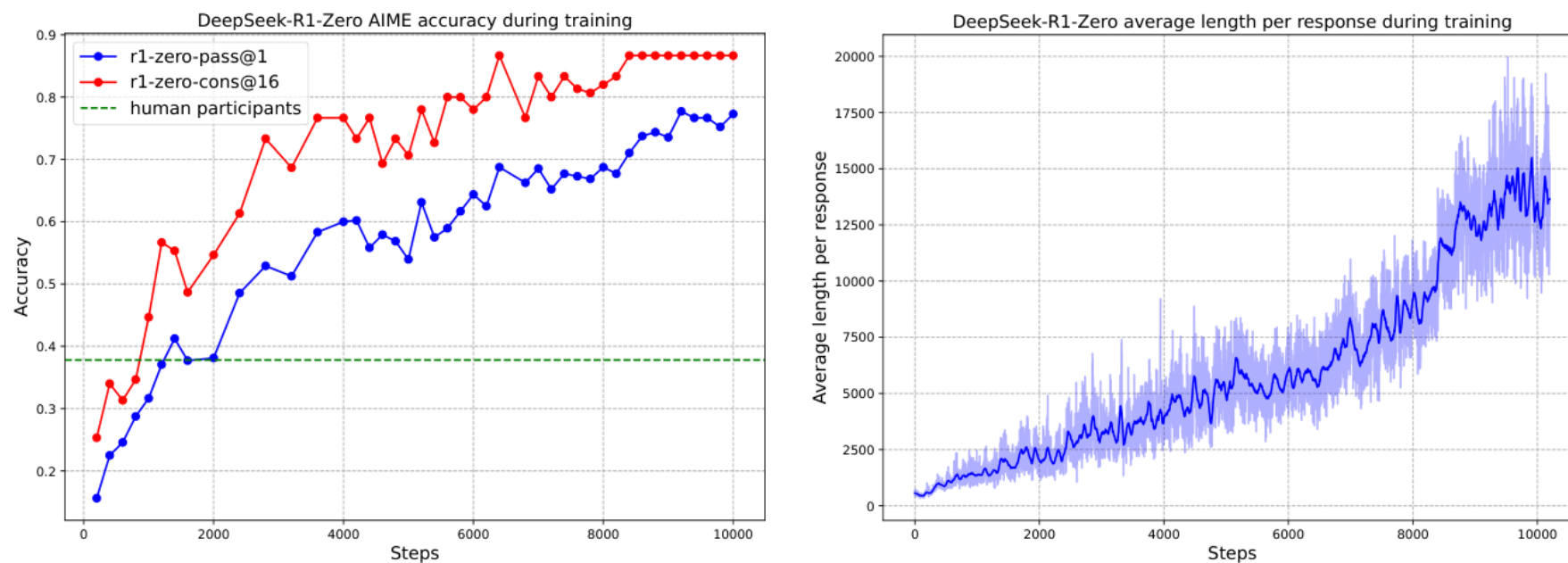


Figure 1 | (a) AIME accuracy of DeepSeek-R1-Zero during training. AIME takes a mathematical problem as input and a number as output, illustrated in Table 32. Pass@1 and Cons@16 are described in Supplementary D.1. The baseline is the average score achieved by human participants in the AIME competition. (b) The average response length of DeepSeek-R1-Zero on the training set during the RL process. DeepSeek-R1-Zero naturally learns to solve reasoning tasks with more thinking time. Note that a training step refers to a single policy update operation.

The 5-stage recipe

The R1 recipe — 5 stages



GRPO — the 30-second intuition

1. Ask the model the **same question G times** (e.g. $G = 16$)
2. **Score** each answer with simple rules (right? well-formatted?)
3. Each answer's **advantage** = how much better than the group average
4. Nudge the policy $\uparrow$ on better-than-average, $\downarrow$ on worse-than-average

No value net. No human rater. The **group is the baseline**.

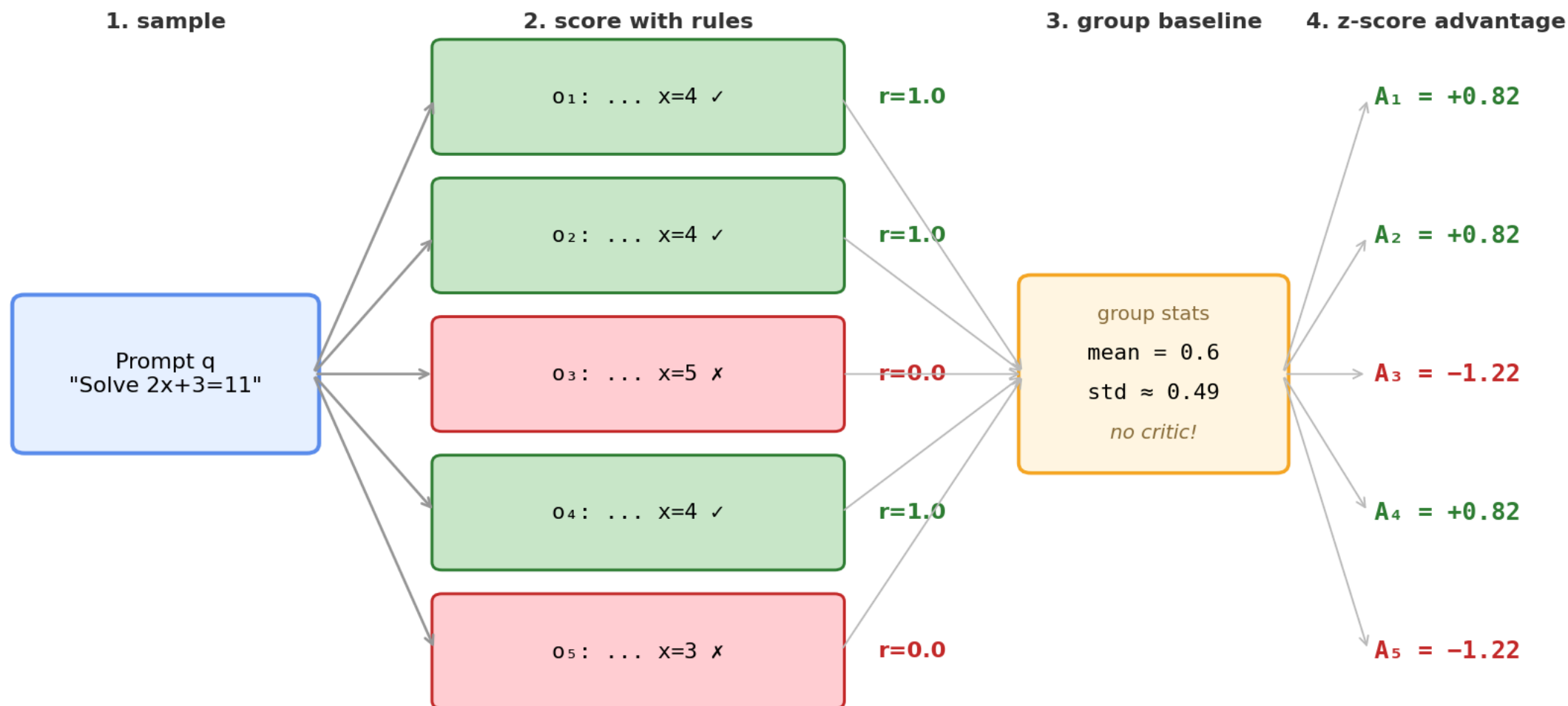
Why no critic?

	PPO (classic RL)	GRPO (DeepSeek)
Baseline	learned value model (extra NN)	mean reward of the group
Networks trained	policy + value	policy only
Memory footprint	~2×	~1×
Reward source	learned reward model (usually)	rule-based (correct? formatted?)

Half the moving parts. No reward hacking from a learned critic.

GRPO in one picture

GRPO in one picture — the group IS the baseline



The formal objective

For each question q , GRPO samples a group of outputs $\{o_1, o_2, \dots, o_G\}$ from the old policy $\pi_{\theta_{old}}$ and then optimizes the policy model π_{θ} by maximizing the following objective:

$$\mathcal{J}_{GRPO}(\theta) = \mathbb{E}[q \sim P(Q), \{o_i\}_{i=1}^G \sim \pi_{\theta_{old}}(O|q)]$$
$$\frac{1}{G} \sum_{i=1}^G \left(\min \left(\frac{\pi_{\theta}(o_i|q)}{\pi_{\theta_{old}}(o_i|q)} A_i, \text{clip} \left(\frac{\pi_{\theta}(o_i|q)}{\pi_{\theta_{old}}(o_i|q)}, 1 - \varepsilon, 1 + \varepsilon \right) A_i \right) - \beta \mathbb{D}_{KL}(\pi_{\theta} || \pi_{ref}) \right), \quad (1)$$

$$\mathbb{D}_{KL}(\pi_{\theta} || \pi_{ref}) = \frac{\pi_{ref}(o_i|q)}{\pi_{\theta}(o_i|q)} - \log \frac{\pi_{ref}(o_i|q)}{\pi_{\theta}(o_i|q)} - 1, \quad (2)$$

where π_{ref} is a reference policy, ε and β are hyper-parameters, and A_i is the advantage, computed using a group of rewards $\{r_1, r_2, \dots, r_G\}$ corresponding to the outputs within each group:

$$A_i = \frac{r_i - \text{mean}(\{r_1, r_2, \dots, r_G\})}{\text{std}(\{r_1, r_2, \dots, r_G\})}. \quad (3)$$

Clipped PPO-style update — but the advantage $\hat{A}$ is just the **z-score of the group rewards**. That's the whole trick.

Reward design

- **Accuracy reward** — does the final answer match?
- **Format reward** — did it use `<think>...</think>` ?

No learned reward model. Rule-based only.

Distillation results

Table 15 | Comparison of DeepSeek-R1 distilled models and other comparable models on reasoning-related benchmarks. Numbers in bold denote the performance is statistically significant (t-test with $p < 0.01$).

Model	AIME 2024		MATH	GPQA Diamond	LiveCode Bench	CodeForces
	pass@1	cons@64	pass@1	pass@1	pass@1	rating
GPT-4o-0513	9.3	13.4	74.6	49.9	32.9	759
Claude-3.5-Sonnet-1022	16.0	26.7	78.3	65.0	38.9	717
DeepSeek-R1-Distill-Qwen-1.5B	28.9	52.7	83.9	33.8	16.9	954
DeepSeek-R1-Distill-Qwen-7B	55.5	83.3	92.8	49.1	37.6	1189
DeepSeek-R1-Distill-Qwen-14B	69.7	80.0	93.9	59.1	53.1	1481
DeepSeek-R1-Distill-Qwen-32B	72.6	83.3	94.3	62.1	57.2	1691
DeepSeek-R1-Distill-Llama-8B	50.4	80.0	89.1	49.0	39.6	1205
DeepSeek-R1-Distill-Llama-70B	70.0	86.7	94.5	65.2	57.5	1633

Limitations

Language mixing, sensitivity to few-shot, weaker on general chat.

Next →

We rebuild every stage in code.

```
presentation/03-r1-recipe-five-stages.md
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